CarePlus: A Privacy-First AI & IoT Healthcare Companion for Elderly Home Care in Nepal

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SUNWAY COLLEGE KATHMANDU — CREATING AI LEADERS

IN ACADEMIC PARTNERSHIP WITH BIRMINGHAM CITY University

SUNWAY RAIN Research & Incubation Center

MOTIVATION

Nepal's ageing population faces a critical care gap as youth emigrate abroad. Existing digital health tools are **English-only, cloud-dependent,** and focused on hospital booking, not ongoing home care.

- Elderly patients miss medicines; emotional states go unmonitored.
- Remote guardians rely entirely on daily phone calls.
- Inconsistent internet makes cloud services unreliable in rural Nepal.
- No affordable, Nepali-language AI companion exists for this demographic.

INTRODUCTION

- **Goal:** A privacy-first, local-AI healthcare companion for elderly Nepali speakers at home.
- **Built:** Raspberry Pi edge device running Whisper STT, Gemma LLM, and Piper TTS, paired with a Bun + Express cloud backend, mobile app, and web dashboard.
- **Team:** Team Git Force — four students at Sunway College Kathmandu with expertise in AI, IoT, and human-centred design.
- **Approach:** Design Thinking + Agile across four sprints.

WHO USES CARE+ — USE CASE DIAGRAM

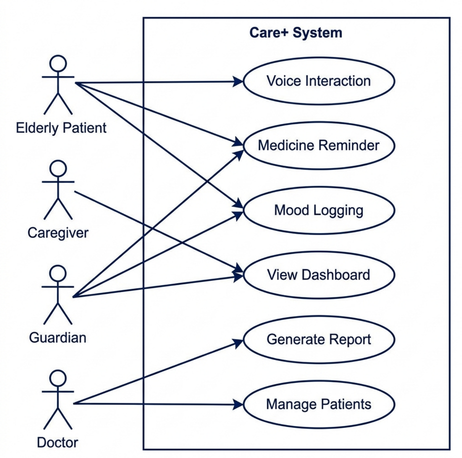

Fig. I: Four actors — Elderly Patient, Caregiver, Guardian, and Doctor — drive Care+'s voice interaction, medicine reminders, mood logging, and clinical dashboards.

SDG ALIGNMENT

Care+ directly supports three UN Sustainable Development Goals:

Primary SDG — *Innovation* — *Equity*

Improves **medication adherence** & wellbeing monitoring (SDG 3); applies edge-AI innovation for under-served communities (SDG 9); bridges Nepal's digital health divide (SDG 10).

SYSTEM ARCHITECTURE

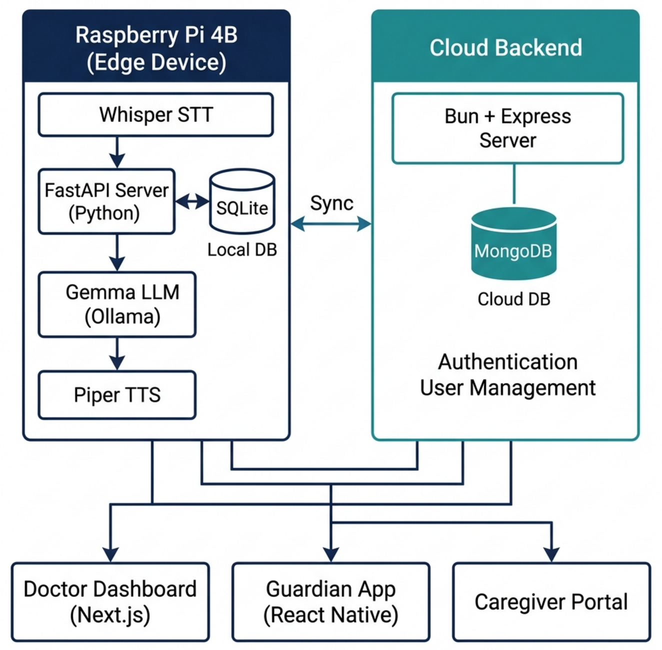

Fig. II: FastAPI + SQLite on Raspberry Pi edge; Bun + Express + MongoDB for cloud auth and user management.

Component	Technology
Edge Device	Raspberry Pi 4B
AI / LLM	Ollama — Gemma 4 (e2b)
Speech-to-Text	Whisper (Nepali fine-tuned)
Text-to-Speech	Piper TTS (Nepali models)
AI API Server	FastAPI (Python 3.10+)
Local Database	SQLite (on FastAPI server)
Cloud Backend	Bun + Express (TypeScript)
Cloud Database	MongoDB Atlas
Mobile App	React Native (Expo)
Web Dashboard	Next.js

Fig. III: Technology Stack

CORE INTERACTION LOOP

1. User speaks naturally in **Nepali or English**
2. **Whisper STT** converts speech to text on-device
3. **FastAPI** orchestrates request to local LLM
4. **Gemma + RAG** determines intent: reply / medicine / mood
5. **Piper TTS** synthesises a natural Nepali speech response
6. Events persisted to **SQLite** & synced to **MongoDB** via Bun

Fig. IV: Core Interaction Loop

MEET CARE+

A friendly on-device voice companion that speaks Nepali, remembers every medicine, and quietly reassures the whole family — **no cloud, no typing, no language barrier.**

APP SCREENSHOTS — LIVE MVP

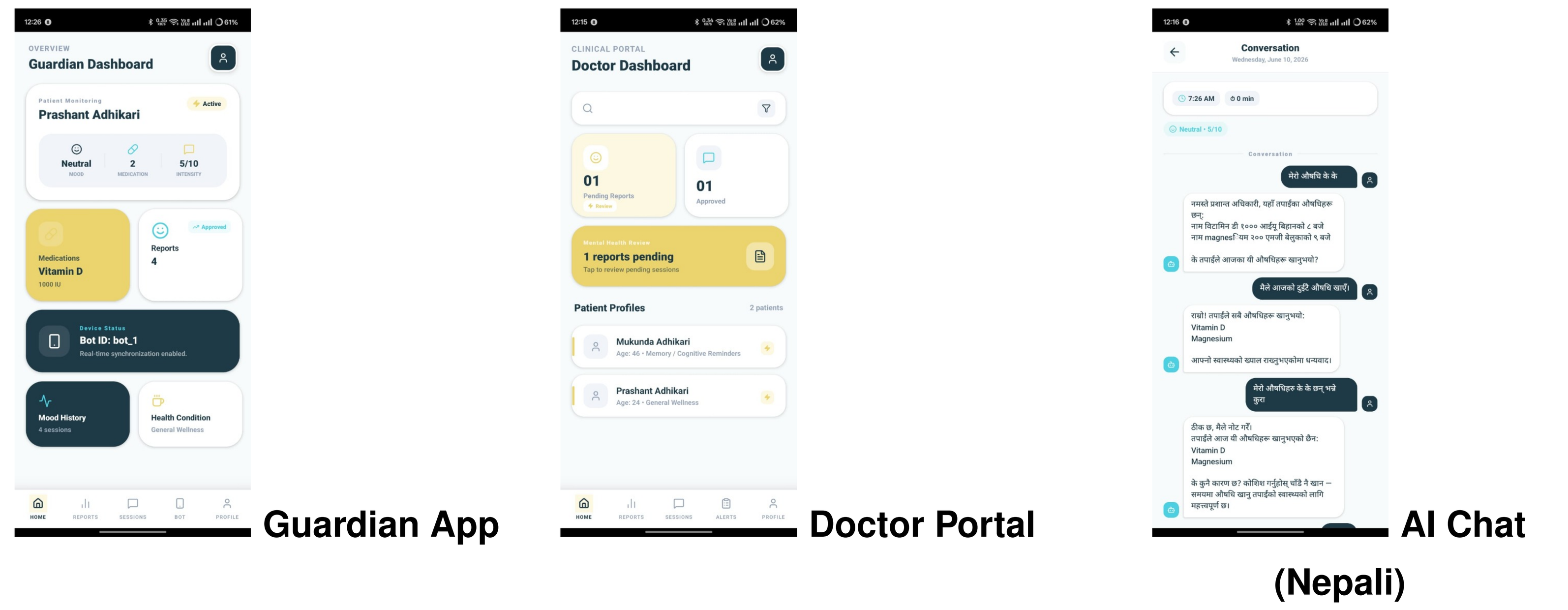

Fig. V: Guardian Dashboard, Doctor Clinical Portal, and Nepali-language AI Conversation.

RESULTS & TESTING

Validated through unit testing, hardware reliability testing, voice recognition across environments, and informal user-acceptance testing with elderly pilot participants.

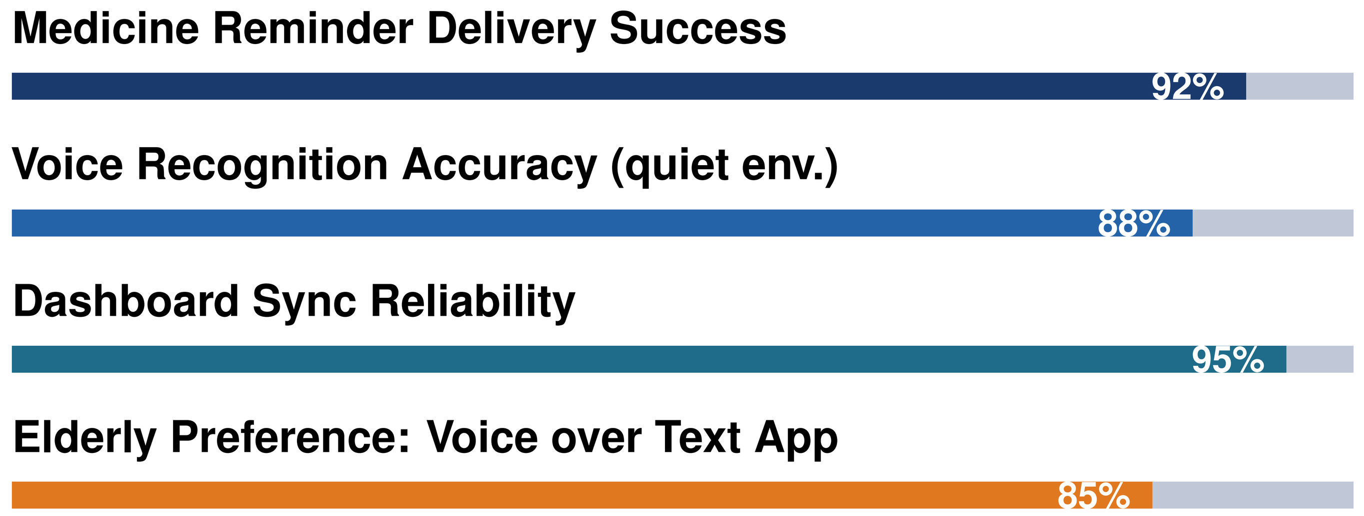

Fig. VI: MVP Performance Indicators (Informal Pilot)

METHODOLOGY — AGILE MILESTONES

M1 **AI Orchestration** *Whisper + Piper + LLM server routing*	**M2** **Medical Mgmt.** *Medicine CRUD, intake confirmation*
M3 **Emotional Analytics** *Mood tracking & PDF reports*	**M4** **Multi-User Ecosystem** *RPi hardening & role dashboards*

CONCLUSION & FUTURE SCOPE

- Care+ proves a culturally relevant AI health companion is buildable within an academic timeframe.
- Local AI eliminates cloud dependency — critical for Nepal's connectivity landscape.
- Voice was significantly more accessible to elderly users than text-based apps.
- **Future:** Wearable integration, clinical validation, expanded South Asian language support, Kathmandu Valley pilot.

REFERENCES

[1] ADI. *World Alzheimer Report 2023.* [2] Radford et al. *Whisper*, ICML 2023. [3] Nepal CBS. *National Census 2021.* [4] WHO. *Ageing & Health*, 2022. [5] Rhasspy. *Piper TTS.* github.com/rhasspy/piper.